

Hamed Shirzad

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Ph.D. candidate in Computer Science at UBC and Student Researcher at Google Research, working on reinforcement-learning post-training of large language models via self-play and self-improvement. First-author publications at NeurIPS, ICML, and AISTATS, and collaborative research in *Nature Biotechnology*. Research experience at Google Research, Valence Labs (Recursion), Borealis AI, Autodesk AI Lab, and the Max Planck Institute. Seeking Research Scientist roles in frontier AI and machine learning for drug discovery.

RESEARCH INTERESTS

Post-training of large language models (reinforcement learning, self-play, self-improvement); sparse and efficient transformers; graph machine learning; generative modeling; self-supervised and representation learning; reliable evaluation of learned representations; machine learning for scientific and drug discovery.

EDUCATION

University of British Columbia

Ph.D. in Computer Science — Advisor: Prof. Danica Sutherland
Focus: efficient transformers, graph learning, and reliable evaluation.

2021 – Present

Vancouver, Canada

Simon Fraser University

M.Sc. in Computing Science — Advisor: Prof. Greg Mori
Thesis: *Graph Generation Using Tree Decomposition*.

2019 – 2021

Burnaby, Canada

Sharif University of Technology

B.Sc. in Computer Engineering & B.Sc. in Mathematics (double major)

2014 – 2019

Tehran, Iran

RESEARCH EXPERIENCE

Student Researcher — Google Research

Remote

Feb. 2026 – Present

- Developing reinforcement-learning post-training methods in which language models self-improve through self-play, generating and learning from their own training signal.
- Fine-tune Gemma 4 models up to 31B parameters with reinforcement-learning fine-tuning techniques, implemented in JAX, Flax, and the Tunix post-training library.

Research Intern — Valence Labs (Recursion), Virtual Cell

Montreal, Canada

Mar. 2025 – Dec. 2025

- Co-developed knowledge-graph-based structured-prediction models of transcriptomic perturbation effects, contributing to TxPert (*Nature Biotechnology*, 2026).
- Developed SeedER, a reinforcement-learning-based seed-and-expand method for retrieval from biological knowledge graphs; first-author paper under review at NeurIPS 2026.
- Collaborated in an interdisciplinary team connecting machine-learning methods with large-scale biological prediction problems in drug discovery.

Research Intern — Borealis AI (now RBC Borealis)

Vancouver, Canada

Sep. 2023 – Feb. 2024

- Developed conditional diffusion models as self-supervised learning backbones for irregular event-sequence and time-series data; first-author paper at the Time Series for Health Workshop, ICLR 2024.

Research Intern — Autodesk AI Lab

Remote

Apr. 2021 – Sep. 2021

- Developed contrastively-learned evaluation metrics for graph generative models, resulting in a first-author NeurIPS 2022 paper.

Research Intern — Borealis AI

Remote

Apr. 2020 – Mar. 2021

- Developed permutation-invariant graph generation via tree decomposition with Prof. Greg Mori and Dr. Hossein Hajimirsadeghi, leading to the first-author AISTATS 2022 paper TD-Gen.

- Worked on kernelized Stein discrepancy and its applications under Dr. Krikamol Muandet.

SELECTED PUBLICATIONS

(* equal contribution; full list on [Google Scholar](#))

SeedER: Seed-and-Expand Retrieval from Knowledge Graphs

H. Shirzad, F. Wenkel, D. Beaini, D. J. Sutherland, E. Noutahi — *Under review, NeurIPS 2026*

Reinforcement-learning-based graph-expansion strategies for knowledge-graph retrieval. [\[arXiv\]](#)

TxPert: Using Multiple Knowledge Graphs for Prediction of Transcriptomic Perturbation Effects

F. Wenkel, W. Tu, C. Masschelein, H. Shirzad, et al. — *Nature Biotechnology, 2026*

Structured prediction over multiple knowledge graphs for biological perturbation modeling. [\[paper\]](#) [\[GitHub\]](#) [\[blog\]](#)

Even Sparser Graph Transformers

H. Shirzad, H. Lin, B. Venkatachalam, A. Velingker, D. Woodruff, D. J. Sutherland — *NeurIPS 2024*

Sparse graph-transformer architecture for scalable learning on graphs. [\[arXiv\]](#) [\[GitHub\]](#)

Exphormer: Sparse Transformers for Graphs

H. Shirzad*, A. Velingker*, B. Venkatachalam*, D. J. Sutherland, A. K. Sinop — *ICML 2023*

Linear-complexity graph transformers via expander graphs; featured on the [Google Research blog](#). [\[arXiv\]](#) [\[GitHub\]](#)

Evaluating Graph Generative Models with Contrastively Learned Features

H. Shirzad, K. Hassani, D. J. Sutherland — *NeurIPS 2022*

Evaluation methodology for graph generative models using contrastively trained representations. [\[arXiv\]](#) [\[GitHub\]](#)

TD-Gen: Graph Generation Using Tree Decomposition

H. Shirzad, H. Hajimirsadeghi, A. H. Abdi, G. Mori — *AISTATS 2022*

Tree-decomposition-based, permutation-invariant graph generation. [\[arXiv\]](#)

A Theory for Compressibility of Graph Transformers for Transductive Learning

H. Shirzad, H. Lin, B. Venkatachalam, A. Velingker, D. Woodruff, D. J. Sutherland — *MLComp Workshop, NeurIPS 2024*

Theoretical analysis of graph-transformer compressibility. [\[arXiv\]](#)

Conditional Diffusion Models as a Self-supervised Backbone for Irregular Time Series

H. Shirzad, R. Deng, H. Zhao, F. Tung — *Time Series for Health Workshop, ICLR 2024*

Diffusion-based representation learning for irregular time series. [\[paper\]](#)

TECHNICAL SKILLS

Languages: Python, C++ , Bash, Java, MATLAB.

ML stack: JAX, Flax, TUnix, PyTorch, PyTorch Geometric, DGL, NumPy, SciPy, pandas, NetworkX.

Methods: LLM post-training, reinforcement learning (RL fine-tuning, self-play), transformers, graph learning, generative and diffusion models, self-supervised learning, model evaluation.

AWARDS & RECOGNITION

2024	Top Reviewer Award, Learning on Graphs (LoG).
2024	Fully Funded Trainee, Advanced Machine Learning Training Network (AML-TN); \$32K stipend and \$4K travel support.
2019	Graduate Fellowship, Simon Fraser University.
2014–17	National Elites Foundation Grant for top university entrance exam achievers and National Olympiad medalists.
2013	Bronze Medal, Iranian National Olympiad in Informatics.

INVITED TALKS

2025	Keynote: <i>Scaling Graph Transformers with Sparse and Sparsified Attention</i> , GSP 2025.
2025	Tutorial: <i>Geometric and Topological Representation Learning</i> , MLSP 2025.

LEADERSHIP & SERVICE

- Reviewer (2022–2026): ICLR, NeurIPS, LoG, AISTATS, TMLR, and JMLR.
- Mentor, Research Experience Program (REX), UBC (2024): guided undergraduate students through a first research project.
- Co-founder, Homa Foundation (2023): helped raise \$45K to support travel for 25 Iranian graduate students relocating to Canada.